

DISCRETE MATHEMATICS

EXERCISES — GRAPH THEORY (SUPPLEMENTARY)

Notation. K_n denotes the complete graph on n vertices; $K_{m,n}$ the complete bipartite graph; C_n the cycle; P_n the path; W_n the *wheel graph* (a hub vertex joined to every vertex of C_n , giving $n + 1$ vertices in total). For a graph G on n vertices: $\delta(G)$ minimum degree, $\Delta(G)$ maximum degree, $\chi(G)$ chromatic number, $\alpha(G)$ independence number, $\omega(G)$ clique number.

Exercise 1

Chromatic number. The *chromatic number* $\chi(G)$ is the minimum number of colours needed to colour the vertices of G so that no two adjacent vertices share a colour.

Determine $\chi(G)$ for each of the following families, distinguishing subcases where necessary. Justify every answer by exhibiting an explicit colouring and arguing no fewer colours suffice.

- (a) K_n for $n \geq 1$.
- (b) C_n for $n \geq 3$, treating even and odd n separately.
- (c) $K_{m,n}$ for $m, n \geq 1$.
- (d) W_n for $n \geq 3$, treating even and odd n separately.

Exercise 2

Greedy colouring bound. Prove that for any simple graph G ,

$$\chi(G) \leq \Delta(G) + 1.$$

(*Hint.* Order the vertices $v_1, \dots, v_n$ arbitrarily. Colour each vertex v_i with the smallest positive integer not already used by any previously coloured neighbour of v_i , and bound the number of colours that can be forbidden at each step.)

Exercise 3

Hamiltonian cycles. A *Hamiltonian cycle* in a graph G is a cycle that visits every vertex exactly once.

- (a) Show that K_n has a Hamiltonian cycle for every $n \geq 3$.
- (b) Show that $K_{m,n}$ with $m \neq n$ has *no* Hamiltonian cycle. (*Hint.* In any cycle on a bipartite graph, the two parts of the bipartition must be visited alternately.)
- (c) Show that $K_{n,n}$ *does* have a Hamiltonian cycle for every $n \geq 2$ by constructing one explicitly.
- (d) **Dirac's theorem** states: if G is a simple graph on $n \geq 3$ vertices with $\delta(G) \geq n/2$, then G has a Hamiltonian cycle. Verify that the hypothesis is satisfied for K_n and for $K_{n,n}$, and check that it fails for $K_{2,3}$.

Exercise 4

Euler's formula and the non-planarity of K_5 . A graph is *planar* if it can be drawn in the plane with no edge crossings. For a connected planar graph, Euler's formula states

$$V - E + F = 2,$$

where F is the number of faces (including the unbounded outer face).

- (a) Draw a planar embedding of K_4 and verify that $V - E + F = 2$ by counting faces.
- (b) Prove that every simple connected planar graph satisfies $E \leq 3V - 6$. (*Hint.* Each face is bounded by at least 3 edges, and each edge borders at most 2 faces; combine this with Euler's formula.)
- (c) Use part (b) to prove that K_5 is not planar.

Exercise 5

Non-planarity of $K_{3,3}$.

- (a) Prove that every simple connected *bipartite* planar graph satisfies $E \leq 2V - 4$. (*Hint.* In a bipartite graph every cycle has even length, so every face is bounded by at least 4 edges.)
- (b) Use part (a) to prove that $K_{3,3}$ is not planar.

Exercise 6

Spanning trees and Cayley's formula. A *spanning tree* of a connected graph G is a subgraph that is a tree and contains every vertex of G .

- (a) Find (by enumeration) all spanning trees of K_3 . How many are there?
- (b) Find (by enumeration) all spanning trees of K_4 . How many are there?
- (c) **Cayley's formula** states that the number of labelled spanning trees of K_n is n^{n-2} . Verify this for $n = 3$ and $n = 4$, and state the result for $n = 5$.
- (d) A spanning tree of K_n is chosen uniformly at random. For $n = 4$, how many spanning trees are paths P_4 ? What fraction of all spanning trees does this represent?

Exercise 7

Independence number and clique number. A set $S \subseteq V(G)$ is *independent* if no two vertices in S are adjacent; $\alpha(G)$ is the size of the largest independent set. A set $S \subseteq V(G)$ is a *clique* if every two vertices in S are adjacent; $\omega(G)$ is the size of the largest clique.

Determine $\alpha(G)$ and $\omega(G)$ for:

- (a) K_n

- (b) C_5
- (c) $K_{m,n}$ (with $m \leq n$)
- (d) W_5 (hub joined to C_5)

Exercise 8

Lower bounds on the chromatic number.

- (a) Prove that $\chi(G) \geq \omega(G)$ for any graph G .
- (b) Prove that $\chi(G) \geq \left\lceil \frac{n}{\alpha(G)} \right\rceil$, where $n = |V(G)|$. (*Hint.* Each colour class is an independent set, so has size at most $\alpha(G)$.)
- (c) Use parts (a) and (b) to compute a lower bound for $\chi(C_5)$, and combine it with Exercise 1(b) to determine $\chi(C_5)$ exactly.

Exercise 9

Bridges and cut vertices. An edge e of a connected graph G is a *bridge* if $G - e$ (the graph with e removed) is disconnected. A vertex v is a *cut vertex* if $G - v$ (the graph with v and all its incident edges removed) is disconnected.

- (a) For the graph G with vertex set $\{1, 2, 3, 4, 5, 6\}$ and edge set $\{12, 13, 23, 34, 45, 46, 56\}$, identify all bridges and all cut vertices.
- (b) Prove that every edge of a tree is a bridge.
- (c) Prove that every internal vertex (degree ≥ 2) of a path P_n is a cut vertex.
- (d) Give an example of a 2-connected graph (connected, no cut vertices) that contains a bridge, or explain why no such graph exists.

Exercise 10

Chromatic number 2 and bipartiteness. Prove the following equivalence for a graph G with at least one edge:

$$G \text{ is bipartite} \iff \chi(G) \leq 2.$$

- (a) Prove ($\Rightarrow$): if G is bipartite with parts A and B , show that the 2-colouring assigning colour 1 to A and colour 2 to B is a proper colouring.
- (b) Prove ($\Leftarrow$): if G admits a proper 2-colouring, show that G is bipartite by exhibiting an explicit bipartition.
- (c) Conclude that $\chi(G) = 1$ if and only if G has no edges, and $\chi(G) = 2$ if and only if G is bipartite with at least one edge.